United States Patent Application Publication	20250284982
Kind Code	A1
Publication Date	September 11, 2025
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Method for Generating a Training Dataset

Abstract

A computer-implemented method for generating a training dataset for training an artificial intelligence for operating a measuring device, in particular a wall diagnostic device, includes (i) receiving unclassified sensor data of at least one sensor unit of a measuring device by a classifier module, (ii) classifying the unclassified sensor data and providing classified sensor data by running the unclassified sensor data through the classifier module, and (iii) adding the classified sensor data to a training dataset.

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Family ID: **1000008505294**

Appl. No.: **19/071380**

Filed: **March 05, 2025**

Foreign Application Priority Data

DE	10 2024 202 234.9	Mar. 11, 2024

Publication Classification

Int. Cl.: **G06N5/022** (20230101)

U.S. Cl.:

CPC **G06N5/022** (20130101);

Background/Summary

[0001] This application claims priority under 35 U.S.C. § 119 to application no. DE 10 2024 202 234.9, filed on Mar. 11, 2024 in Germany, the disclosure of which is incorporated herein by reference in its entirety.
[0002] The present disclosure relates to a method for generating a training dataset for training an artificial intelligence for a wall diagnostic device.
BACKGROUND
[0003] Diagnostic devices for diagnosing walls and detecting objects formed in the walls are known in the prior art.
[0004] It is a task of the present disclosure to provide an improved method for generating a training dataset for training an artificial intelligence of a wall diagnostic device.
[0005] The task is solved by the process set forth below. Advantageous embodiments are subject-matter set forth below.
SUMMARY
[0006] According to one aspect, a computer-implemented method is provided for generating a training dataset for training an artificial intelligence to operate a measuring device, in particular a wall diagnostic device comprising: [0007] receiving unclassified sensor data of at least one sensor unit of a measuring device via a classifier module, wherein the sensor data depicts a wall to be diagnosed with an object of an object type disposed in the wall in an object position; [0008] classifying the unclassified sensor data and providing classified sensor data by running the unclassified sensor data through a classifier module, wherein the classifier module is embodied and configured as an artificial intelligence to perform object recognition based on unclassified sensor data and to determine the object position and/or object type of the object and classify the unclassified sensor data with respect to the object position and/or object type; and adding the classified sensor data to a training dataset.
[0009] This may achieve the technical advantage of providing an improved method for generating a training dataset for training an artificial intelligence to operate a measuring device, in particular a wall diagnostic device. For this purpose, classified sensor data is generated by a classifier module based on unclassified sensor data and added to a training dataset. The sensor data, unclassified as well as classified, here comprises a wall to be examined by the measuring device, including at least one object of an object type disposed in the wall in an object position.
[0010] The classification of the unclassified sensor data by the classifier module is carried out in relation to the object position and/or the object type of the objects disposed in the wall. In that the classifier module is embodied as an artificial intelligence and is configured to perform an object recognition of the objects disposed in the wall depicted by the sensor data based on unclassified sensor data, including determining an object position and/or object type, and to perform a classification of the unclassified sensor data with respect to the object position and/or the object type based on the object recognition, an automatic classification of unclassified sensor data may be achieved. Classified sensor data, in the sense of the application, is labeled sensor data as is known from the field of machine learning.
[0011] According to one embodiment, the unclassified sensor data comprises two-dimensional unclassified radar data, wherein the classifier module was trained, based on the unclassified radar data and taking into account additional classification information, to generate radar data classified in relation to the object position and/or the object type, wherein the classified sensor data comprises the classified radar data and depicts a wall having an object formed in the wall with respect to two spatial dimensions, and wherein the additional classification information comprises information regarding the object position and/or the object type with respect to a third spatial dimension.

[0012] This may achieve the technical advantage of achieving improved object recognition by the classifier module as well as, connected to this, improved classification of the unclassified sensor data by the classifier module. By training the classifier module on classified sensor data, reliable object recognition can be achieved by the classifier module. By considering additional classification information, the object recognition by the classifier module and the classification of the unclassified sensor data based thereon can be further improved.
[0013] According to one embodiment, the additional classification information was generated by performing running the classified two-dimensional radar data through a further artificial intelligence, and wherein the further artificial intelligence is trained to determine the object position and/or object type with respect to the three spatial dimensions based on the classified two-dimensional radar data.
[0014] By doing so, this may achieve the technical advantage that, by using the further artificial intelligence to generate the additional classification information, meaningful additional classification information may be provided that enables the training of the classification module to be performed.
[0015] According to one embodiment, the two spatial dimensions of the classified and/or unclassified two-dimensional radar data are defined by first and second directions perpendicular to each other and parallel to a surface of the wall depicted by the radar data, wherein the third spatial dimension is given by a third direction perpendicular to the first and second directions and directed into the wall, wherein the classified and/or unclassified radar data describes data from a plurality of scanning operations of the measuring device that run side-by-side along the second direction and along the first direction, wherein, in the scanning operations, the measuring device is moved along the first direction relative to the wall and radar data is captured, and wherein the radar data comprises information regarding a signal strength along the third direction directed into the wall.
[0016] This may achieve the technical advantage of the radar data depicting the wall to be examined in three spatial dimensions. The scanning operations of the measuring device are, in the sense of the application, a measurement by the measuring device while the measuring device is moving along the first direction relative to the wall. As the measuring device moves relative to the wall, radar data for the wall is captured. The radar data here comprises signal information in a third direction directed into the wall. By arranging a plurality of scanning operations along the second spatial direction, three-dimensional information for the wall to be examined can be achieved.
[0017] According to one embodiment, classification comprises: [0018] determining the object position along the first direction based on the signal strength information along the third direction, wherein the object position is defined as a position along the first direction with maximum signal strength along the third direction.
[0019] This may achieve the technical advantage of enabling a precise determination of the object position along the first direction. By interpreting the object position along the first direction as the position with maximum signal strength along the third direction, precise object detection based on the radar data may be achieved.
[0020] According to one embodiment, the additional classification information comprises the information regarding signal strength along the third direction.
[0021] This may achieve the technical advantage of providing meaningful additional classification information.
[0022] According to one embodiment, the classified and/or unclassified radar data is illustrated as two-dimensional surface plots in which the data of the plurality of scanning operations are summarized and wherein the determination of the object position is performed jointly for the plurality of scanning operations.
[0023] This may achieve the technical advantage of enabling simple processing of radar data by the classifier module.
[0024] According to one embodiment, the classifier module is configured as a neural network.

[0025] This may achieve the technical advantage of providing a more powerful and reliable classifier module.
[0026] According to one embodiment, the classification of the sensor data is further performed with respect to an object depth and/or an object extent of the object.
[0027] This may achieve the technical advantage of enabling additional information regarding further features of the objects disposed in the walls to be examined to be considered in the training dataset, in the training of the diagnostic module and thus in the wall diagnostics of the diagnostic module.
[0028] According to one embodiment, object classes of the object type of the object comprise: Metal/non-metal object, low voltage cable, single phase AC signal cable, multi phase AC signal cable, wood beam, metal beam, plastic pipe, water filled plastic pipe, for example fresh water pipe, non-water filled plastic pipe, for example waste water pipe, and/or wherein the wall type classes of the wall type of the wall comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or bricks of the wall, floor heating, wall heating.
[0029] This may achieve the technical advantage that a large number of different objects of different object types can be detected or classified. The measuring device or the diagnostic module can be trained hereby to detect and classify common objects installed in building walls. This allows for a particularly precise wall diagnostics in which the detected objects can be precisely and unambiguously assigned to the corresponding object classes.
[0030] By precisely classifying objects and providing the corresponding classification information to the user via the display unit, the most meaningful wall diagnostics possible is enabled. Because the user not only knows that an object is located within the wall and where it is, but also which object type the detected object is, the user can decide accordingly how to carry out further processing of the wall with respect to the detected object. Providing the object types of the object classification of the detected objects thus represents an essential area of the wall diagnostics, because the user can adjust the planned processing of the wall based on the indicated object type.
[0031] According to one aspect, a training dataset for training an artificial intelligence of a wall diagnostic measuring device is provided, wherein the training dataset was generated by the method for generating a training dataset for training an artificial intelligence for operating a measuring device according to one of the preceding embodiments.
[0032] Provided according to one aspect, is a computing unit configured to perform the method of generating a training dataset for training an artificial intelligence for operating a measuring device according to one of the preceding embodiments and/or the method of training an artificial intelligence of a measuring device.
[0033] Provided according to one aspect is a computer program product comprising instructions which, when the program is executed by a data processing unit, prompt it to perform the method for generating a training dataset for training an artificial intelligence to operate a measuring device according to one of the preceding embodiments, and/or the method for training an artificial intelligence of a measuring device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0034] Embodiments of the disclosure are described with reference to the following figures. The figures show:
[0035] FIG. **1** a schematic illustration of a measuring device according to one embodiment;
[0036] FIG. **2** a further schematic illustration of the measuring device according to a further embodiment;
[0037] FIG. **3** a further schematic illustration of the measuring device according to a further

embodiment;
[0038] FIG. **4** a schematic illustration of a measurement of the measuring device according to one embodiment,
[0039] FIG. **5** a further schematic illustration of the measuring device according to a further embodiment;
[0040] FIG. **6** a schematic illustration of a system for generating a training dataset according to one embodiment,
[0041] FIG. **7** a flowchart of a method for generating a training dataset according to one embodiment,
[0042] FIG. **8** a graphical representation of a two-dimensional surface plot of radar data,
[0043] FIG. **9** a graphical representation of a two-dimensional surface plot of radar data, and
[0044] FIG. **10** a schematic illustration of a computer program product.
DETAILED DESCRIPTION
[0045] FIG. **1** shows a schematic illustration of a measuring device **100** according to one embodiment.
[0046] The present disclosure relates to a measuring device, in particular to a wall diagnostic device for examining walls **105** to be processed. Wall diagnostic devices are known in the prior art that are used to detect objects disposed in walls. Such devices allow a user to examine walls to be processed to search for objects disposed in the walls in order to be able to perform planned work, for example drilling in walls, based on this such that damage to the objects disposed in the walls can be avoided.
[0047] In the embodiment shown, the measuring device **100** comprises a housing **150** having a handle **152** for grasping of the measuring device **100** by a user, a display unit **111** for displaying diagnostic results **109** of the wall diagnostics, and controls **154** for switching the measuring device **100** to various operating modes.
[0048] According to the disclosure, the measuring device **100** comprises at least one radar sensor unit **101**. By way of the radar sensor unit **101**, radar signals may be transmitted towards the wall **105** to be examined and radar signals reflected by the wall **105** may be received.
[0049] For example, the radar sensor unit **101** may be configured as a narrow band radar detector device in the 2.4 GHz to 2.4835 GHz frequency range or as an ultra-wide band radar detector device in the 1.8 GHz to 5.8 GHz frequency range.
[0050] The measuring device **100** further comprises a diagnostic module **107** executable on a computing unit **151** of the measuring device **100** to perform the wall diagnostics. The diagnostic module **107** is configured to perform a corresponding diagnosis of the wall to be examined based on the radar data **103** of the radar sensor unit **101**. The radar data **103** of the radar sensor unit **101** thereby depicts the wall **105** to be examined and, if applicable, objects **113** disposed within the wall **105**.
[0051] The wall diagnostics carried out by the diagnostic module **107** comprises at least performing an object recognition. The object recognition here comprises an object detection and an object classification of the object **113** disposed in the wall **105**. The object detection comprises at least the determination of an object position **115**. The object position here describes the positioning of the object disposed in the wall **105** with respect to a reference system defined by the measuring device **100**. The object classification of the detected object **113** comprises at least determining an object type **117** of the detected object **113**.
[0052] The diagnostic results of the wall diagnostics determined in this way, i.e. at least the determined object position **115** and/or the determined object type **117** of the object **113** disposed in the wall **105**, are subsequently presented in a display unit **111** of the measuring device **100** to a user of the measuring device **100**. For example, the display unit **111** may be configured as a corresponding display and the diagnostic results **109** may be visually displayed. Additionally, the display of the diagnostic results **109** may be supported via audible and/or haptic signals. For

example, the haptic signals may be realized via corresponding vibration signals.
[0053] The object **113** can be shown in the display, for example, by way of a corresponding icon. The object **113** can be shown in the corresponding object position **115** in the display. The object extension **121** may be visualized by a corresponding size of the displayed icon. The particular object type **117** of the object **113** may be visualized with a corresponding term or color highlighting of the icon, or by a specific shape of the icon representing the object **113**.
[0054] Alternatively, the wall diagnostics may additionally comprise determining a wall type **123** in the form of a wall type classification of the wall **105** to be examined. The wall type **123** describes the respective type of the wall **105** to be examined. For example, the wall type may be associated with corresponding wall type classes, which may comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or wall bricks, underfloor heating, wall heating or similar wall types found in buildings.
[0055] According to one embodiment, the diagnostic module **107** is further configured to determine an object depth **119** of the object **113** within the wall **105** based on the radar data **103**. The object depth **119** is defined by a distance of the object formed in the wall **105** to a surface of the wall **105**. The distance may be defined on the object side, for example with respect to an object surface or with respect to an object center point. The distance to the surface of the wall **105** describes a shortest distance defined by a direction perpendicular to the surface of the wall **105**.
[0056] According to one embodiment, the diagnostic module **107** is further configured to determine an object extension **121** of the object **113** in at least one predefined direction based on the radar data **103**. The object extension **121** of the object **113** describes a spatial extension of the object **113** in at least one spatial direction, preferably in two spatial directions, particularly preferably in three spatial directions. The object **113** may be described here as a one-dimensional, two-dimensional, or three-dimensional object **113**.
[0057] In conventional use, the measuring device **100** is placed on the surface of the wall **105** to be examined. Radar signals are transmitted towards the wall **105** and radar signals reflected from the wall **105** or the objects **113** disposed there are received via the radar sensor unit **101**. This radar data **103** of the radar sensor unit **101** is used by the diagnostic module **107** to perform the wall diagnostics described above and to determine corresponding diagnostic results **109**.
[0058] For example, diagnostic results **109** may comprise the object position **115** and/or object type **117** of the object **113** disposed in the wall **105**. Alternatively or additionally, the diagnostic results **109** may comprise the wall type **123** of the wall **105** and/or the object depth **119** and/or the object extension **121** of the object **113**.
[0059] The diagnostic results **109** configured in this manner may subsequently be displayed to a user of the measuring device **100** in a display unit **111** of the measuring device **100**. The display unit **111** can be configured as a corresponding display, for example. The diagnostic results **109** may be displayed in graphical or textual form in the display unit **111**.
[0060] According to one embodiment, the measuring device **100** further comprises a motion detection unit **141**. The motion detection unit **141** may be used to detect movement of the measuring device **100** relative to the wall **105**. The motion detection unit **141** may comprise, for example, at least one roller element for this purpose. When the roller element is placed on the wall surface of the wall **105**, movement of the measuring device **100** relative to the wall **105** can be detected when the measuring device **100** moves along a direction of movement **153** by rolling the roller element. Alternatively, the motion detection unit **141** may have a different configuration by which a relative movement of the measuring device **100** relative to the wall **105** can be detected.
[0061] By moving the measuring device **100** relative to the wall **105**, radar data **103** of the radar sensor unit **101** may be captured for a plurality of different positions of the measuring device **100** relative to the wall **105**. This allows a larger spatial area of the wall **105** to be examined than that given by the effective range of the radar sensor unit **101**. This allows for objects **113** to be captured that have a greater spatial extent than the effective range of the radar sensor unit **101**.

[0062] While the measuring device **100** moves along the direction of movement **153**, radar data **103** of the radar sensor unit **101** may be captured continuously. The wall diagnostics may be evaluated by the diagnostic module **107** based on this radar data **103** while the measuring device **100** is moving along the direction of movement **153**. This allows for accelerated wall diagnostics, taking into account the positioning of the measuring device **100** relative to the wall **105**.
[0063] According to its embodiment, the diagnostic module **107** is configured as a correspondingly trained artificial intelligence **125**. The artificial intelligence **125** is trained to perform the above-described wall diagnostics based on the radar data **103** of the radar sensor unit **101** and to determine at least the object position **115** and the object type **117** of an object **113** disposed in the wall **105**. The object classification or determination the object type **117**, respectively, comprises assigning the detected object **113** to predefined object classes.
[0064] The object classes may comprise: Metal/non-metal object, low voltage cable, single phase AC signal cable, multi phase AC signal cable, wood beam, metal beam, plastic pipe, water filled plastic pipe, for example fresh water pipe, non-water filled plastic pipe, for example waste water pipe, or other elements commonly installed in building walls.
[0065] Further, the artificial intelligence **125** may be trained to determine the wall type **123** of the wall **105** to be examined at least based on the radar data **103** of the radar sensor unit **101**. Possible wall types **123** may comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or individual bricks of the brick wall, underfloor heating, wall heating or other similar wall types commonly installed in buildings.
[0066] According to one embodiment, in addition to the radar sensor unit **101**, the measuring device **100** may comprise further additional sensors by way of which additional physical variables are detectable. For example, the measuring device **100** may comprise an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an AC current sensor and/or an NMR sensor and/or an ultrasonic sensor or other sensors commonly used in wall diagnostic devices.
[0067] The diagnostic module **107**, in particular the corresponding trained artificial intelligence **125**, may be configured to perform wall diagnostics based on the radar data **103** of the radar sensor unit **101** and taking into account the additional sensor information of the further sensors described above. The additional information of the additional sensors mentioned above can in particular be used for object recognition of the objects **113** disposed in the walls **105**. The additional sensor information may possibly provide improved detection of the objects **113** and may possibly provide improved classification of the objects **113**.
[0068] In particular, for example, the material of the objects **113**, for example as a metallic or non-metallic material, can be improved and classified by using the additional sensor information.
[0069] FIG. **2** shows another schematic representation of the measuring device **100** according to a further embodiment.
[0070] In the embodiment shown, the measuring device **100** comprises a pre-processing module **127** in addition to the diagnostic module **107**. For wall diagnostics, the measuring device **100** first receives the radar data **103** of the radar sensor unit **101**. Pre-processing of the receiving radar data **103** is performed via the pre-processing module **127**. For example, via pre-processing of the pre-processing module **127**, the radar data may be brought into a corresponding data structure required for wall diagnostics by the diagnostic module **107**.
[0071] As described above, during wall diagnostics, the diagnostic module **107** generates the diagnostic results **109** described above. For example, the diagnostic results **109** may comprise the object position **115** and/or object type **117** and/or object depth **119** and/or object extension **121** of an object **113** disposed in the wall **105** to be examined and/or the wall type **123** of the wall **105** to be examined. The correspondingly generated diagnostic results **109** may subsequently be displayed in the display unit **111** of the measuring device **100**.
[0072] According to one embodiment, in addition to the radar data **103** of the radar sensor unit **101**, the additional sensor information of the additional sensors described above may be included in the

wall diagnostics of diagnostic module **107**. A corresponding pre-processing of the additional sensor information may be performed accordingly by the pre-processing module **127**.

[0073] In the embodiment shown, the diagnostic module **107** comprises a wall type classification module **129** and an object recognition module **131**. The pre-processing module **127** comprises a first pre-processing module **135** and a second pre-processing module **137**. The first pre-processing module **135** comprises an S matrix reduction **155**. The second pre-processing module **137** comprises a background correction **157**, an inverse Fast Fourier transformation **159**, and a focusing and migration **161**. In the pre-processing of the radar data **103** by the pre-processing module **127**, the radar data **103** is first pre-processed by the first pre-processing module **135** and the S-matrix reduction **155** contained therein.

[0074] When doing so, the first pre-processing module **135** generates input data **133** based on the radar data **103**. The input data **133** serves as input data for the wall type classification module **129**. The wall type classification module **129** performs a wall type classification of the wall **105** to be examined based on the input data **133** and generates wall type information **139**. The wall type information **139** contains the wall type **123** of the wall **105** to be examined, as determined in the wall type classification.

[0075] Subsequently, the second pre-processing module **137** performs pre-processing based on the radar data **103** and the wall type information **139**. A background correction **157** of the radar data **103** is performed during this, taking into account the wall type **123** determined in the wall type information **139**. Depending on the wall type **123** of the wall **105** to be examined, different effects on the radar data **103** can occur.

[0076] These effects, which are primarily based on the respective wall type **123** and can affect object recognition, can be corrected by the background correction **157**. After the background correction has been performed, further pre-processing can be carried out by performing the inverse Fast Fourier transformation **159** or focusing and migration **161**, respectively, and input data **133** can be created for the object recognition module **131** once again. Based on the input data **133** provided by the second pre-processing module **137**, the object recognition module **133** performs the object recognition of the object **113** disposed in the wall **105** to be examined and determines at least the object position **115** and the object type **117** of the respective object **113**. Additionally, the object depth **119** and the object extension **121** may be determined by the object recognition module

[0077] According to one embodiment, the diagnostic module is further configured to determine an object depth of the object within the wall based on the radar data, wherein the object depth is defined by a distance of the object formed in the wall to a surface of the wall.

[0078] The pre-processing is optional. Depending on the algorithm used for the diagnostic module **107**, completely unprocessed radar echoes of different frequencies may be used as radar data **103** and as input data for the diagnostic module **107**. Alternatively, radar data **103** processed via multiple steps may be used. The pre-processing steps comprise, for example, the transformation of the signals from the frequency domain to the time or distance domain, background deduction, noise removal, and normalization of the signals. For radar data **103** which is available in the form of complex numbers, only the absolute value can be processed. Alternatively or additionally, the phase information may be considered.

[0079] FIG. **3** shows a further schematic representation of the measuring device **100** according to a further embodiment.

[0080] In the embodiment shown, the diagnostic module **107** comprises a plurality of parallel processing paths **102**. Each processing path **102** includes a pre-processing module **127**, the diagnostic module **107**, for example, comprising the wall type classification module **129** and/or the object recognition module **131** according to the embodiment in FIG. **2**, and a post-processing module **163**.

[0081] In FIG. **3**, primarily the radar data **103** is shown as input data for the wall diagnostics. In addition to the radar data shown, however, the additional information of the additional sensors may also serve as input data for the wall diagnostics. Here, the different information from the different types of sensors in the different parallel processing paths **102** can be processed and the corresponding wall diagnostics performed separately on the different sensor information. Upon completion of the wall diagnostics, a summary module may be used to assemble a summary of the individual sub-analysis results to form the diagnostic results **109** of the wall diagnostics.

[0082] Alternatively or additionally, different sub-aspects of the wall diagnostics may be performed by the different processing paths **102** based on the same sensor information.

[0083] For example, the individual processing paths **102** may process different radar data **103** captured during movement of the measuring device **100** relative to the wall **105** for different positions of the measuring device **100** relative to the wall **105**. The radar data **103**, thus representing different regions of the wall **105** and captured sequentially in time as the measuring device **100** moved relative to the wall **105**, may then be processed in the different processing paths **102** by the modules shown.

[0084] The different processing paths perform a stand-alone wall diagnostics in this respect, comprising at least determining the object position **115** and/or the object type **117** of the object **113** disposed in the wall **105**.

[0085] The summary module **165** may summarize the sub-results provided in the individual processing paths **102** of the stand-alone wall diagnostics of the different regions of the wall **105** into a contiguous diagnostic result **109**. The contiguous diagnostic result here describes the wall diagnostics of a contiguous spatial area that was covered during movement of the measuring device **100** relative to the wall **105** and depicted by the corresponding captured radar data **103**. The parallel processing of the radar data **103** or additional sensor information **104** of the additional sensor elements in the different processing paths **102** thus enables accelerated wall diagnostics.

[0086] Alternatively, various wall diagnostic functions may also be performed in the different processing paths **102**. For example, in one processing path **102**, the wall type classification and the determination of the wall type **123** of the wall **105** to be examined can be performed. In another processing path **102**, object recognition of the object disposed in the wall **113** can be performed. The object detection can be performed with the determination of the object position **115** and the object classification can be performed with the determination of the object type **113** in one processing path **102**.

[0087] Alternatively, the object detection and object classification may then also be performed in two separate processing paths **102**. In further processing paths **102**, the object depth determination, i.e., the determination of the object depth **119** and/or the determination of the object extension **121** may respectively be carried out. In the summary module **165**, the various sub-results of the wall diagnostics may be summarized into corresponding diagnostic results **109**.

[0088] The diagnostic module **107** can be divided into different artificial intelligences **125**, as already shown in the embodiment in FIG. **2**. For example, the diagnostic module **107** may comprise a wall type classification module **129** and an object recognition module **131**. The object recognition module may in turn be divided into an object detection module and an object classification module. The diagnostic module **107** may further comprise an object depth determination module and object extension module, respectively configured to determine the object depth **119** and the object extension **121**.

[0089] The respective modules may each be configured as stand-alone artificial intelligences **125**, for example, neural networks. Alternatively, the various modules may form portions of an overall artificial neural network that are connected to an overall neural network according to structures known in the prior art.

[0090] FIG. **4** shows a schematic illustration of a measurement of the measuring device **100** according to one embodiment.

[0091] For pre-processing, the radar data **103** or the additional sensor information **104** of the remaining sensors may be normalized, in particular to numerically stabilize the subsequent steps performed by the diagnostic module **107** during wall diagnostics. For example, an amplitude and/or offset compensation may be performed for this purpose. Further, to reduce interference, filtering of the radar data **103** may be carried out, and to reduce the data rate the corresponding sensor data may be sampled. Further, the radar data **103** or the additional sensor information **104** may be transformed to the respective required frequency range or time range. Methods known from the prior art can be used for this purpose.

[0092] Further, the captured radar data **103** or additional sensor information **104** may be divided into temporal or spatial windows **167**. Temporal windows **167** may be generated by recording the radar data **103** or the additional sensor information or the pre-processed radar data **103** over a fixed time interval. Spatial windows **167**, on the other hand, may be generated from a mapping of the radar data **103** or additional sensor information **104** to positions of the measuring device **100** relative to the wall **105** along the direction of movement **153**.

[0093] Graph a) of FIG. **4** shows such a data matrix resulting from the steps described above. The data matrix of the window **167** shown in graph a) shows a plurality of sensor data, which may comprise, for example, radar data **103** or additional sensor information **104** of the further sensors plotted along a frequency channel axis **171** or along a space/time axis **169**, respectively.

[0094] A width of the temporal windows **167** may be selected such that different sampling rates of the sensors may be balanced and a new window **167** may be provided frequently enough so that a display of the diagnostic results **109** of the wall diagnostics in the display unit **111** may be shown without too great of a time delay while the measurement is being performed or shortly after the measurement of the measuring device **100** ends.

[0095] A rate of 2 to 20 windows per second of the data recording of the sensor data may be advantageous for this purpose. For spatial windows, the spatial sampling rates can be selected to achieve the desired local accuracy. Advantageously, 1 mm to 1 cm sampling rates can be used. This means that corresponding sensor data is captured every 1 mm to 1 cm of movement of the measuring device **100** along the direction of movement **153**.

[0096] A width of the spatial windows **167** may be selected such that contiguous information relating to an object **113** is contained in a window. Advantageously, a width of the respective spatial windows **167** can be 1 cm to 20 cm. This results in 4 to 100 measured values per window **167**. This allows for further efficient algorithmic processing of the correspondingly recorded radar data **103** or additional sensor information by the diagnostic module **107**.

[0097] A further temporal window **167** or spatial window **167** can be provided as soon as one or more scan points are available.

[0098] The diagnostic module **107** may be oriented such that a matrix corresponding to the window size of the respective spatial or temporal window **167** may be included as input data, for example of each processing path **102** of the embodiment in FIG. **3** as well. According to the embodiment of FIG. **2**, the corresponding input data can comprise the respective pre-processed sensor data, i.e. radar data **103** and additional sensor information **104** of the additional sensors.

[0099] As stated above, wall diagnostics may be performed by the diagnostic module **107** based on a correspondingly trained artificial intelligence. Alternatively, different processing paths may also be calculated by rule-based algorithms. A combination of artificial intelligence and a rule-based algorithm is also possible within a processing path **102** in the form of a parallel connection or concatenation.

[0100] The diagnostic results **109** of the wall diagnostics may be implemented as numeric values, vectors, or matrices. Further, the probability of detection, or for the wall type classification and/or the object classification, respectively, a probability of the specified object classes and/or wall type classifications can be given for the object detection. The same may apply to the position and/or depth determination, for which corresponding probability values can also be given.

[0101] If, in addition to the radar data **103**, the additional sensor information of the further sensor types is processed in a processing path **102**, these may either be merged within the artificial

intelligence **125** or combined by rule-based combinations.
[0102] In the post-processing of each processing path **102**, of the embodiment of FIG. **3**, multiple algorithm results based on multiple windows **167** may be summarized by the summary module **165**. This summary can in particular be realized by majority formation, sum formation or also by multiplication of successive probability values.
[0103] Further, by clustering multiple results, for example, it is possible to detect which objects of multiple detected objects lying close to one another are the same object so that they are not incorrectly detected multiple times.
[0104] Likewise, it is possible to multiply a weighting function **177** when summarizing the results from multiple windows **167**. Advantageously, the diagnostic sub-results **175** corresponding to corresponding data points in the space may be weighted with respect to positioning of the diagnostic sub-results **175** relative to a center point of the respective window **167**. This is illustrated by way of example in graph b), in which the individual diagnostic sub-results **175** are weighted according to the weighting function **177** shown with respect to the center point of the window **167** shown.
[0105] According to one embodiment, the results of one processing path **102** after post-processing **163** may influence the extension of another processing path **102**. Here, weighting parameters may be adjusted that may depend on the particular result from the processing path **102** for each window.
[0106] For example, the result of an object classification in which the object type **117** of an object disposed in the wall **105** is defined can be utilized to increase the weight of a wall type classification in which the wall type **123** of the respective wall **105** is determined in the post-processing at locations without objects **113**, because the respective radar data **103** at these locations are less influenced by reflections of the objects **113**.
[0107] FIG. **5** shows a further schematic representation of the measuring device **100** according to a further embodiment.
[0108] Graphs a) and b) of FIG. **5** show two different alternatives of a joint data processing of radar data **103** and additional sensor information **104** by the diagnostic module **107**.
[0109] Graph b) illustrates a joint processing of the radar data **103** and the additional sensor information **104** of the additional sensors by the diagnostic module **107**. For this purpose, the radar data **103** and the additional sensor information **104** are collectively used as input data of the diagnostic module **107** configured as an artificial intelligence, in particular as an artificial neural network. The diagnostic module **107** here comprises multiple convolutions **108** and multiple dense layers **106**. The radar data **103** and the additional sensor information **104** are processed jointly as input data via the convolutions **108** and dense layers **106**. The aforementioned diagnostic results **109** are created as output data of the diagnostic module **107**, based on this.
[0110] In graph b), in contrast, the radar data **103** and the additional sensor information **104** are used as stand-alone input data of the diagnostic module **107**. The diagnostic module **107** becomes multiple processing paths **102**. The processing paths **102** each comprise multiple convolutions **108** and at least one dense layer **106**. In the various processing paths **102**, wall diagnostics are performed separately by the diagnostic module **107** based on the radar data **103** and the additional sensor information **104**, respectively.
[0111] In an additional concatenation layer **148**, the partial results of the partial diagnoses of the different processing paths **102** are combined and fed to a final dense layer **106**. The output data of the diagnostic module **107** corresponds to the diagnostic results **109** described above.
[0112] The correspondingly configured diagnostic module **107** is designed to perform wall diagnostics as described above, including the features described above, based on the radar data **103** and the additional sensor information **104**.
[0113] In the embodiment shown, the diagnostic module **107** is configured as an artificial neural network, in particular as a convolutional network. Corresponding network architectures with convolutions **108**, dense layers **106**, and concatenation layers **148** are known in the prior art.

[0114] FIG. **6** shows a schematic representation of a system **600** for generating a training dataset **143** according to one embodiment.

[0115] In the embodiment shown, the system for generating a training dataset **143** comprises at least one classifier module **183**. The classifier module **183** is embodied as a trained artificial intelligence and is configured to generate classified sensor data **174** for the training dataset **143** based on unclassified sensor data **172**. The classifier module **183** is trained to perform object recognition of objects **113** disposed in the wall **105** depicted by the sensor data **172** based on the unclassified sensor data **172**. The object recognition comprises an object detection with determination of the object position **115** and/or an object classification with determination of the object type **117** of the respective object **113**.

[0116] In the embodiment in graph b), the system **600** furthermore comprises a further artificial intelligence **132** in addition to the classifier module **183**. The further artificial intelligence **132** is trained to determine additional classification information **130** based on classified radar data **128**. The classified radar data **128** here depicts walls **105** and objects **113** disposed in them to be examined. The classified radar data **128** is thereby classified at least in relation to the object position **115** and/or the object type **117** of the objects **113** disposed in the depicted walls **105**. The classified radar data **128** are in two-dimensional form and comprise a plurality of scanning operations of the measuring device **100** that run along the first spatial direction, wherein the plurality of scanning operations are disposed adjacent one another along a second direction. The first and second directions run along the X and Y-axis of the coordinate system of FIG. **1** and are thus arranged parallel to the surface of the wall **105** to be examined.

[0117] The further artificial intelligence **132** is trained, despite application to the two-dimensional radar data **128**, to determine the object position **115** of the objects **113** disposed in the wall **105** in three spatial directions X, Y, Z. The additional classification information **130** generated by the further artificial intelligence **132** describes the object position **115** and/or the object type **117** with respect to the third spatial direction. The third spatial direction is aligned along the Z-axis of the coordinate system shown in FIG. **1** and thus runs into the depicted wall **105**.

[0118] The further artificial intelligence **132** is trained to determine the object recognition of the objects **113** disposed in the walls **105** depicted by the radar data **128** in all three spatial directions based on the classified radar data **128**. The further artificial intelligence **132** is thus at least configured to determine the object position **115** of the objects **113** with respect to the three spatial directions X, Y, Z of the coordinate system shown in FIG. **1** based on the two-dimensional classified radar data **128**.

[0119] The classifier module **183** is now trained based on unclassified radar data **126**, taking into account the additional classification information **130** provided by the further artificial intelligence **132**. The unclassified radar data **126**, in turn, is in two-dimensional form and comprises a plurality of radar data captured in a plurality of scanning operations **140**. The scanning operations **140** describe measurements of the measuring device **100** in which the measuring device **100** is moved along the first spatial direction X relative to the wall to be examined **105**, during which it captures radar data **103**. The measurements of the scanning operations are performed for various locations on the wall **105** disposed along the second spatial direction Y. This makes it possible to examine a two-dimensional spatial area of the wall **105** and map it using the correspondingly captured radar darts. For this purpose, the data captured in the plurality of different scanning operations **140** are disposed adjacent to each other along the second spatial direction Y and summarized into the two-dimensional radar data.

[0120] Based on the two-dimensional unclassified radar data **126** and in consideration of the additional classification information **130**, the classifier module Z is trained according to prior art training methods to generate classified radar data **128**. The radar data classified **128** by the correspondingly trained classifier module **183** is thereby classified at least in relation to the object position **115** and/or the object type **117** of the objects **113**.

[0121] The classified radar data **128** used in the graph d) for training the further artificial intelligence **132** and the classified radar data **128** generated during training of the classifier module **183** need not be the same radar data but may instead represent different data.
[0122] The unclassified radar data **126** used for training the classifier module **183** may have been generated by a plurality of different measurements of a plurality of different measuring devices **100**.
[0123] FIG. **7** shows a flowchart of a method **300** for generating a training dataset **143** according to one embodiment.
[0124] To generate a training dataset **143**, in a first method step **301**, unclassified sensor data **172** of at least one sensor unit of a measuring device **100** is initially received by a classifier module **183**. The sensor data **172** depicts the wall **105** to be diagnosed and the objects **113** disposed therein. The sensor data **172** may comprise radar data **103** and additional sensor information **104**.
[0125] In a further method step **303**, the unclassified sensor data **172** is classified by the classifier module **183** and classified sensor data **174** is generated. The classifier module **183** is embodied as appropriately trained artificial intelligence and is configured to perform an object recognition based on unclassified sensor data **172** and to determine the object position **115** and/or the object type **117** of the objects **113**. The classifier module **183** is configured to perform a corresponding classification of the unclassified sensor data **172** with respect to the object position **115** and/or the object type **117** of the objects **113** based on the object recognition.
[0126] According to one embodiment, the unclassified sensor data **172** is two-dimensional unclassified radar data **126**. The classifier module **183** is trained on the unclassified radar data to generate classified radar data **128** with respect to the object position **115** and/or the object type **117**. The unclassified radar data **126** and the classified radar data **128** comprise two-dimensional radar data, and depict the wall **105** to be examined with respect to a first spatial direction X and a second spatial direction Y arranged perpendicular thereto. The first and second spatial directions X, Y are parallel to a surface of the wall **105**. The additional classification information **130** comprises information regarding the object position **115** and/or the object type **117** in relation to a third spatial direction Z perpendicular to the first and second spatial directions X, Y.
[0127] According to one embodiment, the classification information **130** was generated by running the classified two-dimensional radar data **128** through a further artificial intelligence **132**. The further artificial intelligence **132** is trained to determine the object position **115** and/or the object type **117** relative to the three spatial dimensions X, Y, Z based on the classified two-dimensional radar data **128**.
[0128] The classified and/or unclassified radar data **126**, **128** comprises data of a plurality of scanning operations **140** of the measuring device **100** that run side-by-side along the second direction Y and extend along the first direction X. The scanning operations **140** of the measuring device **100** here describe measurements of the measuring device **100** while moving the measuring device **100** along the first direction X, wherein corresponding radar data **103** is recorded during the measurement. The radar data **103** comprises information regarding a signal strength along the third spatial direction directed into the wall **105**.
[0129] In another method step **307**, the classification **303** comprises determining the object position **115** along the first direction X based on the signal strength information along the third direction Z, wherein the object position **115** is defined as a position along the first direction X with maximum signal strength along the third direction Z.
[0130] In another method step **305**, the sensor data **174** classified by the classifier module **183** is merged into the training dataset **143**.
[0131] FIG. **8** shows a graphical representation of a two-dimensional surface plot **142** of radar data **103**.
[0132] FIG. **8** shows a surface plot **142** of two-dimensional radar data **103**. The surface plot **142** shows radar data with respect to the first spatial direction X that runs parallel to the surface of the

wall **105** to be examined and runs along the third spatial direction Z that is pointing into the wall. The surface plot **142** further shows a measurement signal **146**. The measurement signal **146** is disposed along the first spatial direction X. A signal strength of the measurement signal **146** is plotted along the third spatial direction Z. Further, the surface plot **142** shows an envelope of the signal strength **144**.
[0133] According to one embodiment, the object position **115** of an object **113** disposed in the wall **105** depicted by the radar data of the surface plot **142** may be determined along the first spatial direction X based on the signal strength of the measurement signal **146** in the third spatial direction Z. The object position **115** of the object **113** represented by the measurement signal **146** along the first spatial direction X can thus be identified as the position along the first spatial direction X in which the signal strength of the measurement signal **146** assumes a maximum value with respect to the third spatial direction Z. The maximum value is represented by the peak of the envelope **144**. The respective object position **115** with respect to the first direction X is represented by the X symbol within the surface plot **142**.
[0134] FIG. **9** shows a graphical representation of a two-dimensional surface plot **142** of radar data **103**.
[0135] FIG. **9** shows a further surface plot **142**. The surface plot **142** shows radar data extending along the first spatial direction X and the second spatial direction Y. The surface plot **142** in turn shows a measurement signal **146**.
[0136] The surface plot **142** is formed by a plurality of scanning operations **140** disposed adjacent one another along the second spatial direction Y. The scanning operations **140** extend along the first spatial direction X. Thus, to create the surface plot **142**, radar data is captured while carrying out multiple scanning operations **140** by moving the measuring device **100** along the first direction X and capturing radar data **103** during this time. To generate a plurality of scanning operations **140**, the measuring device **100** is placed at various positions along the second spatial direction Y. Starting from these positions, the measuring device **100** is moved along the first spatial direction X to record the data of the scanning operations **140**.
[0137] FIG. **10** shows a schematic representation of a computer program product **500** comprising instructions that, when the program is executed by a data processing unit, cause the latter to perform the method **300** for generating a training data set **143**.
[0138] In the embodiment shown, the computer program product **500** is stored on a storage medium **501**. The storage medium **501** can in this case be any desired storage medium known from the prior art.

Claims

1. A computer-implemented method for generating a training dataset for training an artificial intelligence for operating a measuring device, comprising: receiving unclassified sensor data of at least one sensor unit of a measuring device by a classifier module, wherein the sensor data depicts a wall to be diagnosed with an object of an object type disposed in the wall in an object position; classifying the unclassified sensor data and providing classified sensor data by running the unclassified sensor data through the classifier module, wherein the classifier module is embodied and configured as an artificial intelligence to perform an object recognition based on unclassified sensor data and to determine the object position and/or the object type of the object and classify the unclassified sensor data with respect to the object position and/or the object type; and adding the classified sensor data to a training dataset.
2. The method of claim 1, wherein: the unclassified sensor data comprises two-dimensional unclassified radar data, the classifier module was trained based on the unclassified radar data and taking into account additional classification information, to generate radar data classified in relation to the object position and/or the object type, the classified sensor data comprises the classified radar

data and depicts a wall having an object formed in the wall with respect to two spatial dimensions, and the additional classification information comprises information regarding the object position and/or the object type relative to a third spatial dimension.
3. The method of claim 1, wherein: the additional classification information was generated by running the classified two-dimensional radar data through a further artificial intelligence, and the further artificial intelligence is trained to determine the object position and/or the object type relative to the three spatial dimensions based on the classified two-dimensional radar data.
4. The method of claim 3, wherein: the two spatial dimensions of the classified and/or unclassified two-dimensional radar data are defined by first and second directions which are perpendicular to each other and parallel to a surface of the wall depicted by the radar data, the third spatial dimension is given by a third direction perpendicular to the first and second directions and directed into the wall, the classified and/or unclassified radar data describe data of a plurality of scanning operations of the measuring device that run side-by-side and along the first direction and along the second direction, in the scanning operations, the measuring device is moved along the first direction relative to the wall and radar data is captured, and the radar data comprises information regarding a signal strength along the third direction directed into the wall.
5. The method of claim 1, wherein the classifying comprises: determining the object position along the first direction based on the signal strength information along the third direction, wherein the object position is defined as a position along the first direction with maximum signal strength along the third direction.
6. The method of claim 2, wherein the additional classification information comprises the signal strength information along the third direction.
7. The method of claim 1, wherein the classified and/or unclassified radar data is represented as two-dimensional surface plots in which the data of the plurality of scanning operations is summarized, and wherein the determination of the object position is performed jointly for the plurality of scanning operations.
8. The method of claim 1, wherein the classification of the unclassified sensor data is further performed in relation to an object depth and/or an object extension of the object.
9. The method of claim 1, wherein object classes of the object type of the object comprise: metal/non-metal object, low voltage cable, single phase AC signal cable, multi phase AC signal cable, wood beam, metal beam, plastic pipe, water filled plastic pipe, non-water filled plastic pipe, and/or wherein the wall type classes of the wall type of the wall comprise: concrete wall, plasterboard/drywall wall, brick wall and/or bricks of the wall, floor heating, wall heating.
10. A training dataset for training an artificial intelligence of a wall diagnostic measuring device, wherein the training dataset was generated according to the method for generating a training dataset according to claim 1.
11. A computing unit configured to perform the method of generating a training dataset for training an artificial intelligence for operating a measuring device of claim 1.
12. A computer program product comprising instructions which, when the program is executed by a data processing unit, prompt the data processing unit to perform the method for generating a training dataset for training an artificial intelligence to operate a measuring device according to claim 1.
13. The method of claim 1, wherein the measuring device is a wall diagnostic device.
14. The method of claim 9, wherein: the water filled plastic pipe includes a fresh water pipe, and the non-water filled plastic pipe includes a waste water pipe.